

DrainDead: Emptying Batteries of Parked Electric Vehicles

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Abstract

In recent years, sales of electric vehicles have skyrocketed. Fueled by rising gas prices and government incentives, many companies and private car owners have switched from internal combustion engine vehicles to battery electric vehicles. Although bidirectional charging promises to provide power to homes and grid stabilization in the future, it is only rarely used today. This paper discusses the possibilities of discharging batteries from electric vehicles on the road today and how attackers could use this approach to drain energy from targeted cars. We also present a prototype for performing such attacks. Furthermore, this paper includes test results obtained from discharging various electric vehicles from different manufacturers.

1 Introduction

Stealing fuel from parked vehicles, especially trucks, is a widespread crime. According to a 2014 report by the German Federal Office of Goods Transport, fuel theft incidents nearly doubled between 2010 and 2013 in Germany [2]. Criminals often simply break the petrol cap open or drill a hole in gas tanks. This attack is simple to perform, hard to prevent, and causes damage in fuel costs and vehicle repairs. With the ongoing transition from internal combustion engine-powered vehicles towards electric ones, this easy approach to stealing fuel will not be possible in the near future. Although breaking open the charge port is still possible for electric vehicles, the car charge controller could theoretically prevent attackers from draining energy through the charge port. A potential attacker might pose as a bidirectional charging station, asking the vehicle to drain energy.

A charging station certificate is required to discharge a vehicle using the official bidirectional charging standard ISO 15118-20. Similarly to how a browser only trusts a Transport Layer Security (TLS) connection when the server can provide a certificate signed by a trusted certificate authority (CA), bidirectional charging stations require a certificate for communicating with the vehicle. While modern browsers contain

a large list of commonly used CAs, no such list exists in the vehicle ecosystem, resulting in many vehicles only trusting a small number of CAs. As a result, most of the charging sessions performed today do not use encryption but communicate in plaintext. However, this encryption is mandatory for the bidirectional charging standard ISO 15118-20, resulting in most charging sessions today being based on older charging standards without this requirement. Although a potential attacker cannot start a bidirectional charging session, he can communicate with a target vehicle using older communication standards.

This paper will demonstrate how a potential attacker can abuse the basic principles of DC fast charging to drain energy from a vehicle, even when one of the old charging standards is used. Before discussing the implementation details of our attack device, we will first describe the basic procedures normally used when DC is charged. We will also model potential attackers, their interests and capabilities. Lastly, we will discuss the potential impact of this attack and possible remedies.

This paper focuses on the charging standards used in Europe and America. Although other regions use different standards, such as GB/T or CHAdeMO, they are outside the scope of this work.

2 Related Work

We will first examine the widely adopted charging standards DIN SPEC 70121 and ISO 15118-2 to understand how a vehicle is charged. Afterwards, we will briefly examine the bidirectional charging standard ISO 15118-20 and some other attacks on electric vehicle charging equipment that have previously been demonstrated.

2.1 Battery charging standards

When charging an electric vehicle directly from the grid, the car controls the current drawn to match its internal limits. Because of this, the only information an AC charging station

needs to communicate to the vehicle is the maximum current allowed.

DC charging stations, however, are based on external rectifiers that convert AC power from the grid to DC power for the vehicle's battery. This means that the amount of current is directly controlled by the charging station rather than the vehicle. For voltage and current to stay within safety limits of the battery, the vehicle needs to communicate its limits to the charging station. The industry standard ISO 15118 [8] was created for this communication.

As described in detail by Loew et al. [12], this standard is based on more or less common technologies for each step in the communication:

- For the physical layer, a powerline network between the vehicle and the charging station is established as a first step.
- For both vehicle and charging station to join the same powerline network, a custom handshaking protocol "Signal Level Attenuation Characterization" (SLAC) defined in ISO 15118-3 [9] is used.
- IPv6 is used as the network layer, utilizing stateless autoconfiguration for address assignment.
- The vehicle sends out a discovery UDP broadcast, to which the charging station responds, including the address and port the vehicle shall connect to.
- The vehicle connects to the TCP port given in the discovery response. This TCP connection is then used for all further communication.

After an established connection, the ISO 15118-2 standard consists of a list of requests the vehicle can send over the connection to the charging station. For each request, the charging station responds with a corresponding response packet. After some initial packets covering protocol version selection, payment selection, and safety checks, precharging happens. During precharging, the charging station supplies the requested voltage to the vehicle without closing the vehicle's contactors. This step prevents a high inrush current when the battery's main contacts are closed. Afterwards, the majority of the communication happens in the main charge loop. The main charge loop consists of two packets: `CurrentDemandReq` and `CurrentDemandRes`. The former one is sent by the vehicle to request a specific target voltage and charge current. The charging station then responds with the latter one, providing the actual voltage read and the actual current supplied. These two packets are repeated until the charging session is completed or canceled by one of the two parties.

Apart from the widely supported ISO 15118-2 standard today, there exists an older DIN SPEC 70121 fork [1], which was published while ISO 15118-2 was still under development for vehicles already in production at that time. It is identical in

all the aspects mentioned above, except for the actual payload packets, which are named slightly differently and lack some of the functionality introduced with ISO 15118-2.

Although ISO15118-2 theoretically allows TLS to encrypt the TCP communication between the vehicle and the charging station, most of today's charging sessions are done in plaintext. While this may change in the future, both vehicles and charging stations must allow unencrypted communication to remain compatible with older hardware.

2.2 Battery discharging standards

When ISO 15118-2 was introduced, it focused mainly on fast charging vehicles using DC power. Although this is still the main use case for high-level charging communication, the market demanded high-level communication even during AC charging, e.g., limiting charge power based on solar production. Additionally, one major upcoming use case for electric vehicles is to charge and discharge them, delivering energy to power home appliances or even feeding that energy into the grid for stabilization.

The new charging standard ISO 15118-20 was introduced to meet these demands. Its basic communication patterns are similar to the two charging standards described in the previous section, utilizing powerline, SLAC, IPv6, TCP, and EXI. However, while transport security is not available or optional with the other standards, the ISO 15118-20 standard enforces TLSv1.3 to encrypt and secure the communication between the vehicle and the charging station. Both sides need a certificate signed by a mutually trusted certificate authority for this to work. As of today, only a small number of vehicles and charging stations have a certificate installed or even support TLS, so this standard does not play a large role.

Although the standard will likely play a role in future discharge scenarios, it was not considered part of this work to discharge vehicles due to its unavailability.

2.3 Bidirectional Charging Stations

Residential charging stations are devices that are used to connect electric vehicles to the grid. Traditionally, they are only used to charge vehicles, not to discharge them. Although their components are similar to fully featured charging stations, they usually do not come with means for accurate metering and payment. Thus, their primary difference from non-residential charging stations is their use in residential buildings and offices. Today, a small number of companies are developing bidirectional charging stations. In theory, the two options differ in the types of energy provided to the vehicle:

Bidirectional AC charging stations works similarly to unidirectional AC charging stations, where the vehicle's on-board charge controller performs the conversion from AC to DC. The charging station acts as a simple one- or three-phase AC

plug with added communication. As this kind of communication is usually based on ISO 15118-20, they communicate with the vehicle, commanding it to drain or provide power on the AC lines at a specific rate. Thus, these devices have no special electronics regarding bidirectionality but act as a communication bridge between the house and the vehicle.

DC bidirectional charging stations, however, have to convert the DC power from the battery to AC power delivered to the grid and vice versa. Although the required power electronics make DC charging stations significantly more expensive, many vehicles do not support bidirectional AC charging, forcing homeowners to opt for DC instead.

Although some newer models, such as the new electric Renault R5 or the Kia EV9, support bidirectional AC charging, the majority of the models on the road today support bidirectional charging only via a DC or not at all.

2.4 Attacks on Electric Vehicle Charging

In 2022, Köhler et al. introduced and validated the Broken-Wire wireless attack, as detailed in [11]. This attack exploits a vulnerability in the physical layer of the Combined Charging Standard, particularly focusing on the HomePlug Green PHY powerline communication protocol. They performed a Denial-of-Service attack by inundating the communication channel with frame preambles. Per the HomePlug specifications, stations employ a Carrier-sense multiple access with collision avoidance (CSMA/CA) protocol, necessitating channel checking before transmitting. By exploiting this collision avoidance feature, the attack hinders communication between the vehicle and the charging station, thus terminating the charging session and preventing it from resuming. The susceptibility is increased by using single-ended, unshielded cables in powerline communication, allowing the attack to be executed wirelessly at distances of up to 47 meters, thereby amplifying its potential threat.

2.5 Accessing the charge port

For discharge attacks to be conducted, the vehicle's charge port must be accessible. This is usually not the case for parked vehicles that are not charging, as it locks the charge port door as part of the central locking system, similar to regular gasoline cars. We propose three methods to overcome this limitation:

1. Some OEMs implement a comfort feature for their customers to unlock and open the charge port lid from outside the vehicle by sending a special wireless signal to the car, e.g., by pressing a button on the key fob (Hyundai, Nissan, Opel) or specialized equipment at the charging park. An attacker can launch a sniffing and/or replay attack on these wireless systems to open the charge port non-destructively without any owner interaction, if the signal is not authenticated. For Tesla

cars, this attack has been demonstrated by replaying the Sub-1GHz ISM band signal sent out by their super-charger stations that automatically opens the charging port lid by pressing a button on the charging cable (see CVE-2022-27948).

2. The charging port can also be accessed by interrupting an ongoing charging session, which leads to an automatic unlocking of the charging cable. This could be done by triggering an emergency stop button that some charging stations offer, or by using any attacks to disrupt the communication channel between the car and the charging station (e.g. [11]).
3. Finally, the charge port lid can also be opened by physical force, as it is usually not part of the car alarm system. However, this method is destructive and is likely to damage the car.

We acknowledge that disabling the onboard charger when the car is parked could mitigate the attack described in this paper. However, for the vehicles we tested during this research (Figure 3), we verified that the electric vehicle charge controllers keep responding to our requests even if the car was shut down with the charging port lid locked.

3 Threat and Attacker Models

This threat model is based on ISO 15118 [8] and Bao et al. [3], with a focus on securing the EV charging infrastructure and ensuring the safety of vehicle and data exchange in public environments, specifically during the parking and charging phases. The following section outlines the various elements of the threat model.

Environment The vehicle is parked in a public area. During the assessment, it was not charged and was locked, adding a layer of physical security. The vehicle has a combined charging system connector and supports the DIN SPEC 70121 or ISO 15118-2.

Authentication and Data Protection The vehicle might support TLS and ISO 15118-20, requiring the charging station to authenticate itself using a valid certificate issued by a trusted certificate authority. However, since TLS is optional for ISO 15118-2, neither encryption nor authentication is required to start a charging session with the vehicle.

Attacker Model The attacker in this model has several capabilities and characteristics. They have physical access to the vehicle, which could include being near it during the charging process. The attacker does not need to break any encryption, such as TLS or data encryption, because TLS is not mandatory in the charging standards. Therefore, the

attack could be carried out regardless of whether or not encryption is implemented. Additionally, the attacker does not have access to internal documentation, meaning they do not have insider knowledge of the vehicle or the charging infrastructure. They are also well-versed in the cybersecurity landscape, potentially owning specialized hardware for interfacing with vehicle interfaces, which could enable them to exploit known vehicle attacks and related vulnerabilities. The attacker can also create a custom portable DC charging station, as described in Sections 4.2 and 4.3, which can be used as a malicious charging point to interact with the vehicle during charging sessions. Furthermore, the attacker knows the charging protocols, allowing them to exploit the weaknesses between the car and the charging station.

Attack Vectors This scenario has several potential attack vectors. An attacker could try to overcharge or overcurrent the battery or perform software-based attacks on the vehicle's charge controller. This work focuses on discharging the vehicle's battery, allowing the attacker to steal energy from parked cars. A potential attacker might use discharging attacks to sell energy to the grid or use the stolen energy in other vehicles. Furthermore, drained batteries can act as a Denial-of-Service (DoS) attack since the vehicle cannot move afterwards. If the vehicle is left at a low state of charge for extended periods, such attacks could also be used to permanently damage the battery.

Assumptions This threat model assumes that the current EV charging infrastructure and vehicles are not authenticated or adequately secured, leaving the system vulnerable to exploitation. Although secure authentication mechanisms such as digital certificates and signatures are in place, they are not used in practice [13]. As a result, this model accounts for the lack of authentication in specific real-world scenarios, making the system susceptible to attacks based on physical access, communication manipulation, and other vulnerabilities. In summary, this threat model emphasizes the weaknesses in the current EV charging infrastructure, particularly when considering public charging stations and the interaction between vehicles and custom charging points. Although encryption and other security mechanisms are recommended, their limited implementation today leaves the system open to various attacks, including those targeting physical access, communication exploits, and custom charging points.

4 Implementation

The following subsections describe how a vehicle can be discharged without using the official bidirectional charging standard ISO 15118-20. Afterwards, we will describe which components we used to build a device to perform our discharge tests. Since this device has a very high price tag, we will then

describe a second prototype for discharging attacks, which uses off-the-shelf components, to demonstrate the feasibility of discharge attacks even without a big upfront investment. Finally, we present our test procedure on various vehicles that identifies the potential for draining their batteries.

4.1 Discharging vehicles using ISO 15118-2

Bidirectional charging was introduced in ISO 15118-20. In theory, older vehicles that only support -2 or the older DIN SPEC standard and do not support the new -20 standard do not support discharging. With -2 charging, the vehicle reports a voltage and current demand to the charging station, which must then provide the requested voltage. It can provide a current up to the maximum transmitted by the vehicle. This means that, when following the standard, the vehicle cannot signal a maximum discharge current.

However, for a rogue charging station, it is still possible to drain energy because of the electrical details of DC fast charging: After the ISO 15118-2 connection has been established and the initialization steps have been successfully passed, the vehicle directly connects the battery to the DC poles of the charge port, giving the charging station direct access to the battery. This is typically done because the power converters of stationary fast charging stations are more powerful than the power converters included in the car, allowing higher charging power. With direct access to the battery, a rogue charging station can charge and potentially discharge the vehicle's battery.

Although vehicles could theoretically detect this abnormal behavior of a rogue charging station by measuring the direction of current flow, our tests show that today's vehicles do not immediately prevent charging stations from discharging their batteries. Our tests show that some vehicles cancel the charging session after some time or after a significant amount of energy has been drained from the battery. However, the attacker can restart the charging process and continue discharging the battery.

4.2 Design of a portable bidirectional DC charging station

To perform these kinds of discharge attacks, a DC charging stations with programmable current is required, which allows us not only to charge a vehicle but also to discharge it. In general, a bidirectional charging station consists of three major components:

- Communication and control logic handling communication with the vehicle and controlling the charger and inverter components.
- A battery charger for transforming AC energy from the grid to DC energy to charge the battery.

- An inverter for transforming the DC energy from the battery to grid-synchronized AC energy.

For our tests, we used a charging communication evaluation board designed and manufactured by chargebyte [4] to handle the low-level handshake and powerline communication to ethernet conversion. On the software side, we used the open-source ISO 15118 protocol stack developed by EcoG and published on GitHub [7]. We used a bidirectional 1000 V, 3000 W power supply sold by Elektroautomatik for the high-voltage battery charger and inverter. The resulting test device, shown in figure 1, can charge and discharge vehicles. By modifying and extending the open source software stack, we were able to design custom charging procedures, which, after successfully establishing an ISO 15118 charging session and precharging, switches the power supply to sink mode, effectively discharging the vehicle with up to 3000 W.

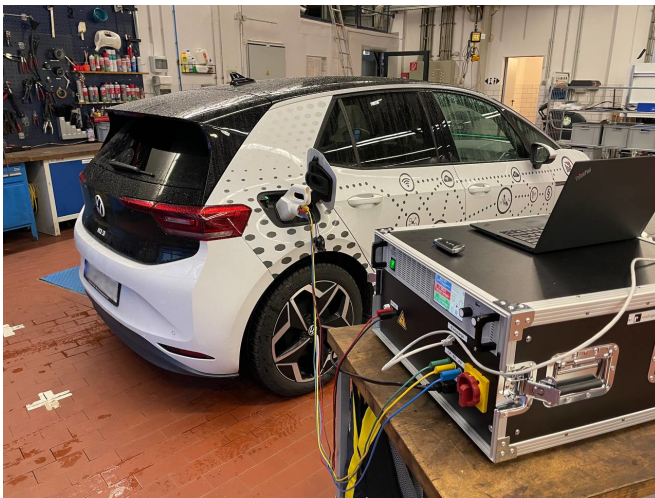


Figure 1: Portable bidirectional DC charging station discharging a Volkswagen ID.3

4.3 Off the Shelf Components for discharging

Since the components used for our portable DC charging station are expensive equipment usually used in product design, they are not readily available to potential attackers. Some of the components are only sold to business customers, so an attacker might not be able to buy all of them. Even if an attacker can source all components, the combined price of around 10,000 € would require many discharge attacks before the initial investment is amortized.

For this reason, we decided to build a second discharging prototype, using only easily accessible and cheap hardware: Since powerline communication is widely used in home networking, many people have already successfully modified off-the-shelf AC HomePlug modems, resoldering them to work

as HomePlug modems for charging communication [14]. The low-level communication can also be easily implemented using a cheap microcontroller. We used an Arduino Nano to generate the PWM and an operational amplifier to convert it into a 12V PWM signal. However, the DC-to-AC conversion is the most expensive component when building a bidirectional charging station. Instead of using an industrial high-voltage power supply, we used a hybrid solar inverter capable of charging and discharging batteries up to 500 V. The open source project “Battery Emulator” [5] uses the same setup for charging and discharging vehicle batteries. Still, instead of using the charge port and ISO 15118, it talks directly to the CAN bus of the car battery in a second-life scenario. We used this open-source project to emulate a BYD Premium battery when communicating with a Fronius solar inverter. We made it accept vehicles connected via CCS and drain power from the battery to power connected loads. These solar inverters are currently available from 900 € (3 kW) to 1800 € (10 kW) and are thus almost five times cheaper than the bidirectional power supply used originally. Together with wiring, our off-the-shelf discharging device costs around 1200 €, significantly cheaper than our portable DC charging station and more affordable than all DC charging stations available today.



Figure 2: Low cost discharging device consisting of a Fronius Gen24 Plus and a CCS adapter box

4.4 Vehicle Compatibility Test Procedure

Several tests per vehicle are required to test the behavior of various vehicles in terms of charging, communication, and discharging. Usually, this boils down to starting a charging ses-

sion with a specific communication configuration and power profile. While usually a charging session is stopped by unplugging the CCS connector from the vehicle, we were able to trigger vehicle unplug detection through software using a modified CCS connector. This allows us to perform any number of charging sessions by simply unplugging and then virtually re-plugging the connector without changing any physical connections.

To test the vehicle's capabilities, we defined various charging communication configurations in combination with different power profiles. Our test setup runs each test for 180 seconds before terminating the charging session and starting the next one. This automation allowed us to run the same tests on different vehicle models from other manufacturers, giving an overview of which vehicle prevents discharging attacks under what circumstances.

The following tests have been performed:

- Using the vehicle's preferred protocol and power parameters
- Using DIN SPEC 70121 only
- Using ISO 15118-2 only
- Using ISO 15118-20 only
- Enforcing transport layer security
- Only accepting plug and charge as payment
- Switching between charging for 20 seconds and discharging for 20 seconds
- Not supplying actual voltage during precharge, but communicating fake voltages to the vehicle

4.5 Tested Vehicle Capabilities

While discharging the vehicle battery was possible with nearly all vehicles we tested, some cars seem to detect discharging and cancel the charging session after some time has passed or some amount of energy has been drained.

For example, all tested Volkswagen Group vehicles cancel the charging session after 60 seconds of discharging. This remains the case when switching between charging and discharging every 20 seconds: The vehicles add up the total discharge time and cancel the charging session after exactly 60 seconds of discharge time.

Other vehicles, such as the Tesla Model 3, only detect discharging when a significant amount of energy has been drained from them. In this case, a potential attacker could restart the charging session and continue discharging. All other vehicles tested do not stop a potential attacker from draining the battery.

Some tested vehicles do not require the charging station to supply an actual precharge voltage. Instead, they trust the

EVSEPresentVoltage value communicated by the charging station.

Our test setup never reached the actual charge loop when trying to test the capabilities of BMW vehicles. The BMW canceled the charging session before exiting the precharge state. We reached out to BMW to find potential issues with the charging communication, but did not receive a response in time.

Figure 3 shows an overview of all vehicles tested and the results for each capability test as described in the previous section. The column "maximum voltage" contains the end-of-charge voltage for each vehicle. Since the IONIQ 6 uses an 800 V battery architecture, it incorporates a DC/DC converter to convert the voltage supplied by the charging station to the battery's actual voltage. This allows the IONIQ 6 to charge at DC charging stations limited to a maximum output of 500V.

5 Impact and Mitigation

Discharging an electric vehicle battery can have a financial impact and perform a denial-of-service attack. This section briefly covers these impacts and also possible mitigations that could be put in place to prevent such attacks in the future.

Financial Impact When energy is discharged from a vehicle, the owner must recharge the vehicle again. Depending on the type of vehicle and its battery size, this can add up to anything between 30 kWh and 400 kWh.

For combustion engine vehicles, the higher financial impact is not the fuel itself, but rather the damage done by an attacker to drain the fuel tank and the indirect damage due to the vehicle's unavailability. This could also apply to electric vehicles; for example, when the attacker uses force to break open the charge port, repair and indirect costs can easily outweigh the costs for recharging the battery.

Denial of Service Utilizing discharge attacks, a vehicle can be completely discharged, leaving the owner unable to drive. When the attacker's goal is not to steal energy but rather to prevent the victim from continuing their journey, a faster and cheaper discharge device could be designed that simply converts the discharged energy to heat.

Mitigations The best approach would be to only allow TLS-secured charging sessions using mutual authentication for both the vehicle and the charging station as defined by ISO 15118-20 [10]. However, as shown by Szakály et al. [13], many charging stations built today do not support ISO 15118-20. Thus, new vehicles will need to support unauthenticated communication to be interoperable with legacy charging stations.

As described in the previous sections, discharging attacks can only be performed when an attacker can access the charge

Vendor	Model	DIN SPEC 70121	ISO 15118-2	TLS	maximum voltage	requires precharge	can be discharged	stops discharging	stops discharging after
Hyundai	IONIQ 6	✓	✓	✓	up to 825.6V	✗	✓	✗	>180s
Hyundai	IONIQ	✓	✗	✗	378.4V	✗	✓	✗	>180s
Kia	e-Niro	✓	✗	✗	421.4V	✗	✓	✗	>180s
Volkswagen	e-Golf	✓	✓	✗	369.0V	✓	✓	✓	60s
Volkswagen	ID.3	✓	✓	✓	464.0V	✓	✓	✓	60s
Volkswagen	ID.4	✓	✓	✓	413.0V	✓	✓	✓	60s
Cupra	Born	✓	✓	✓	464.0V	✓	✓	✓	60s
Opel	Astra-e	✓	✓	✗	430.9V	✓	✓	✗	>180s
Tesla	Model 3	✓	✗	✗	422.0V	✓	✓	✓	15Ah
Ford	Mustang Mach E	✓	✓	✓	414.0V	✓	✓	✗	>180s
Maxus	eDeliver 3	✓	✗	✗	383.4V	✗	✓	✗	>180s
BMW	iX	✓	✓	✗	-	✓	-	-	-
BMW	i5	✓	✓	✗	-	✓	-	-	-

Figure 3: Capability Test Results for various vehicles

port. While an attacker may forcefully open the charge port, the vehicle could refuse to start a charging session when locked.

As shown by our research, at least vehicles based on the Volkswagen MEB platform already employ a mitigation technique to defend against discharge attacks: When connected to a DC charging station without being charged, they cancel the charging session after 60 seconds. Due to the time needed to restart a charging session, especially for initializing the new charging session, this method effectively slows down and thus prevents any realistic attempts towards discharging. This mitigation promises to be an effective measure and, more importantly, since it can be implemented purely in software, it is the recommended approach to prevent unauthenticated discharging of vehicle batteries in the future.

6 Ethical Considerations

In order to perform the demonstrated attack, a potential attacker needs to have physical access to the vehicle and the charge port. Typically, the charge port is mounted behind a lockable cover, similar to gas tanks, preventing a potential attacker from easily accessing the charge port.

As described in 2.5, for Tesla vehicles, the charge port cover can be opened through a wireless attack that is easily available using off-the-shelf sub-GHz penetration testing tools, such as a Flipper Zero. This means discharge attacks could be performed on many Tesla vehicles today. For this reason, we contacted Tesla in January 2025 and reported on the procedure involved in performing this attack. Unfortunately, as of today, we have not received a response from the Tesla security team.

Additionally, we informed all other affected OEMs with general information about the possibility of discharging their vehicles. As physical access to the charge port of an unlocked vehicle is necessary and not typically available, we classify the real-world impact on customers as minimal. Consequently, we have yet to receive any feedback from the OEMs at the time of writing.

7 Open Policy & Data Access

Apart from the hardware described in Section 4.2, we created a DC charging station software stack. Since the free and open source EcoG/iso15118 [7] software stack already implements high-level communication with the vehicle, only hardware abstractions for our power supply and low-level handshaking had to be implemented. The source code of the software used to collect the data for this work is published on GitHub: <https://github.com/securityinmobility/dc-charging-station>

Since partner research institutes own most of the vehicles tested, their identifying information, such as vehicle identification numbers and MAC addresses, is not considered personal

data. Thus, we published pcapng dumps of high-level communication from all vehicles not owned by individuals on GitHub as well: <https://github.com/securityinmobility/vehiclelogs/tree/main/vehicle2grid> By using the free and open source dissector for charging communication “dsV2Gshark” [6], these files can be viewed using Wireshark [15].

8 Conclusion

As this research shows, vehicles on the road today can be charged and discharged even when they do not support bidirectional charging standards. We have classified the behavior of various vehicles of different manufacturers when discharging their batteries, demonstrating how older charging standards can be abused to discharge vehicles. Our demonstrated bidirectional DC charging station can be a valuable tool for cybersecurity researchers who want to test various aspects of Vehicle-to-Grid communication. Additionally, we also discussed how a significantly cheaper bidirectional DC charging station can be built using off-the-shelf components.

In future work, the demonstrated attack could be tested in combination with attacks such as Brokenwire [11] to first stop an ongoing charging session and then unplug the connector and plug in a malicious charging station instead. Future work could also evaluate the use of resistive heating to quickly drain large amounts of energy from a vehicle to further increase the feasibility of discharge attacks for denial-of-service attacks.

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